

Nataly Smith

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ABOUT ME

Senior Software Engineer focused on AI systems, scientific computing, and full-stack development. Background in healthcare, genomics, mathematical modeling, and cloud infrastructure.

SKILLS SUMMARY

- **Languages:** Python, TypeScript/JavaScript, C#, C, C++, Rust, Fortran, SQL, Bash, MATLAB, R
- **AI/ML:** LLMs, Agentic Workflows, LangGraph, PyTorch, Scikit-Learn, OpenCV
- **Frameworks:** .NET, React, Next.js, Flask, Tauri, Flutter, Electron
- **Infrastructure:** AWS, Azure, GCP, Firebase, Docker, PostgreSQL, GitHub Actions, CI/CD

RELEVANT EXPERIENCE

- Senior Software Engineer, AI Platform**
Function Health

Remote
August 2025 - Present

 - Build and maintain AI systems for Function’s health platform.
 - Develop customer-facing LLM applications and agentic workflows.
 - Improve AI platform reliability, performance, and developer tooling.
- Cohort II AI Challenger**
Gauntlet AI

Austin, TX
June 2025 - August 2025

 - Selected for Cohort II of a highly selective AI engineering fellowship requiring a 40+ CCAT score.
 - Built AI applications and agentic systems across weekly engineering projects.
 - Worked with LLMs, RAG, MCP, computer vision, scientific computing, and cloud infrastructure.
- Senior Software Developer**
Bruker Cellular Analysis

Remote
January 2023 - June 2025

 - Completed full-scale Azure DevOps migration in 3 months by coordinating team efforts and creating custom tools.
 - Re-architected a critical application, improving performance by 400% through code optimization.
 - Reduced data processing runtime by 75%, meeting critical product specifications.
 - Built required testing functionality ahead of schedule, expediting product release by 2 weeks.
- Software Developer**
Bruker Cellular Analysis

Remote
January 2022 - January 2023

 - Developed research tools for next-generation sequencing and genomic analysis.
 - Built a web portal that gave interdepartmental teams access to internal research tools.
 - Introduced automated unit and integration testing into CI/CD pipelines.
 - Integrated C++ computer vision libraries into sequencing applications.
- Computer Systems Engineer**
McMaster-Carr

Elmhurst, IL
September 2021 - January 2022

 - Developed internal web applications and integrated legacy systems with modern business applications.
- Directed Research: Miller-Jensen Laboratory**
Yale University

New Haven, CT
January 2020 - May 2021

 - Built computational models of latent HIV activation and analyzed gene-expression data using R and MATLAB.
 - Published findings in PLOS Computational Biology.

SELECTED PROJECTS

- **LAPACK AI:** Implemented AlphaTensor’s 4x4 matrix multiplication algorithm in LAPACK using Fortran, C/C++, and OpenCL GPU acceleration.
- **GeneKnow:** Built a local-first genomic analysis application using Rust, Tauri, React, TypeScript, and LangGraph.

EDUCATION

- Yale University**
Bachelor of Science - Biomedical Engineering; GPA: 3.64

New Haven, CT
August 2017 - May 2021

PUBLICATION & HONORS

- **Publication:** A transcriptional cycling model recapitulates chromatin-dependent features of noisy inducible transcription (PLOS Computational Biology, 2022)
- **Recognition:** Cum Laude, National Hispanic Scholar, Yale CBIT Healthcare Hackathon Runner-Up
- **Languages:** Fluent in English, Spanish, French